📄 CASE STUDY — QueryLens

AI-assisted Search Ads Evaluation Ops Console

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1. Overview

QueryLens is a full-stack MVP designed to simulate how Search Ads Operations teams evaluate ad relevance, combine AI and human judgment, and identify model failure cases.

The goal of this project was to move beyond a simple AI demo and build a **realistic operational workflow tool** that reflects how product, operations, and algorithm teams collaborate to improve search monetization quality.

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2. Problem

Search monetization requires balancing two competing goals:

- maximize commercial revenue
- maintain a high-quality user experience

If ads are irrelevant:

- users lose trust
- engagement drops
- long-term ecosystem health is affected

Operations teams are responsible for:

- reviewing ad relevance at scale
- maintaining consistent labelling standards
- identifying weak areas in ranking or matching logic
- feeding insights back to product and algorithm teams

However, challenges include:

- high review volume
- inconsistent labelling
- lack of prioritization
- limited visibility into model errors

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3. Goal

Design a system that:

- supports **end-to-end ad relevance evaluation**
- uses **AI to accelerate review**
- allows **human override**
- identifies **AI vs human disagreement**
- provides **metrics for continuous improvement**

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4. Solution

I built **QueryLens**, a lightweight evaluation console with three key layers:

1. Review Workflow

- Display search query + ad candidates
- Run AI evaluation per ad
- Assign human relevance label
- Add reviewer notes

2. AI-Assisted Evaluation

The system uses an LLM to generate:

- query intent classification
- relevance label
- relevance score
- explanation

3. Analytics Dashboard

Tracks:

- agreement vs disagreement
- label distribution
- query-level disagreement rates

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5. Product Design Decisions

Human-in-the-loop approach

AI provides a first-pass evaluation, but humans make the final decision.

Disagreement-focused workflow

Instead of reviewing everything equally, the system highlights:

- cases where AI is likely wrong

Simple label taxonomy

Used 3 labels:

- Highly Relevant
- Partially Relevant
- Irrelevant

This reduces ambiguity and improves consistency.

Metrics-driven design

Added dashboards to ensure evaluation results are:

- measurable
- actionable

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6. Key Features

AI Evaluation

- intent detection
- relevance classification
- scoring
- reasoning

Human Review

- override AI label
- add contextual notes
- finalize evaluation

Disagreement Detection

- highlights mismatches between AI and human
- supports error analysis

Dashboard

- agreement rate
- disagreement rate
- label distribution
- query-level performance

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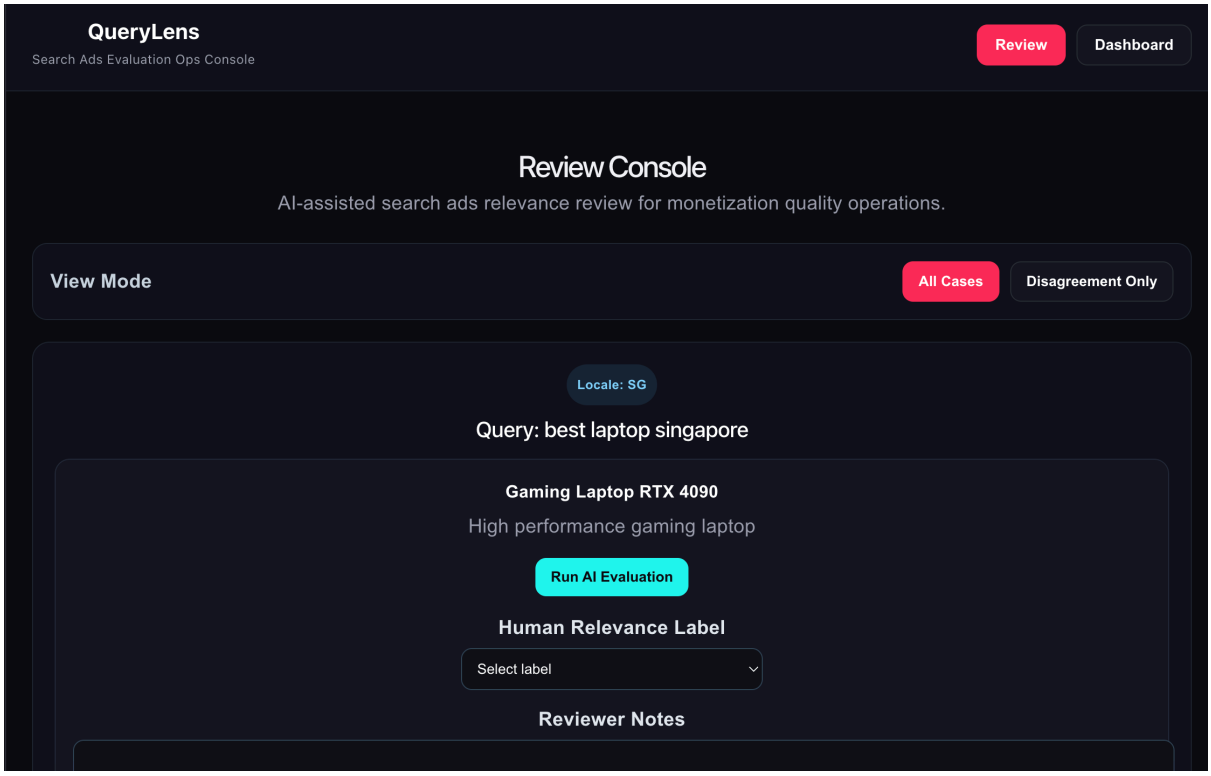
7. Screenshots

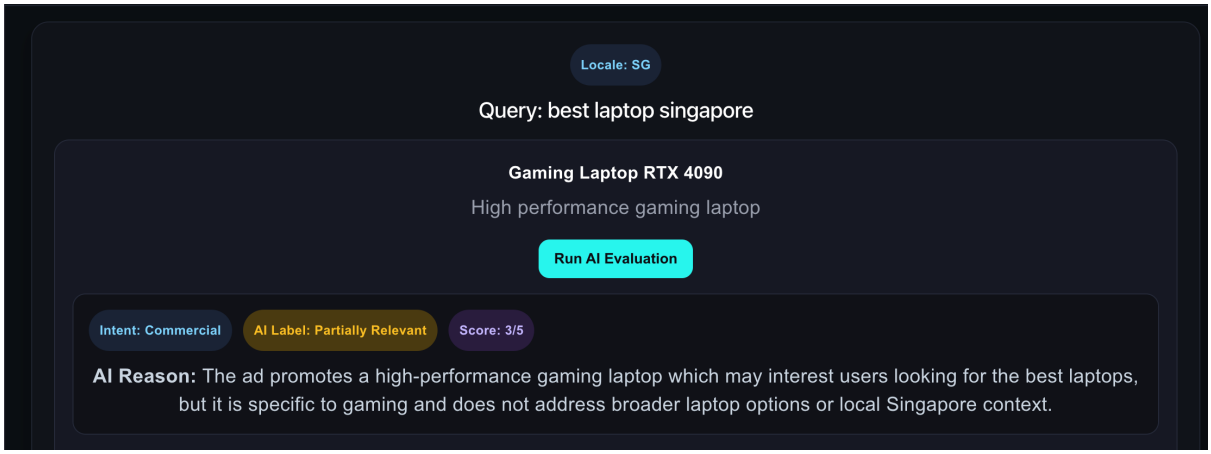
🔷 Review Console (Main Workflow)



Description:

- Displays query and ads
- Shows AI evaluation results
- Allows human labelling and notes

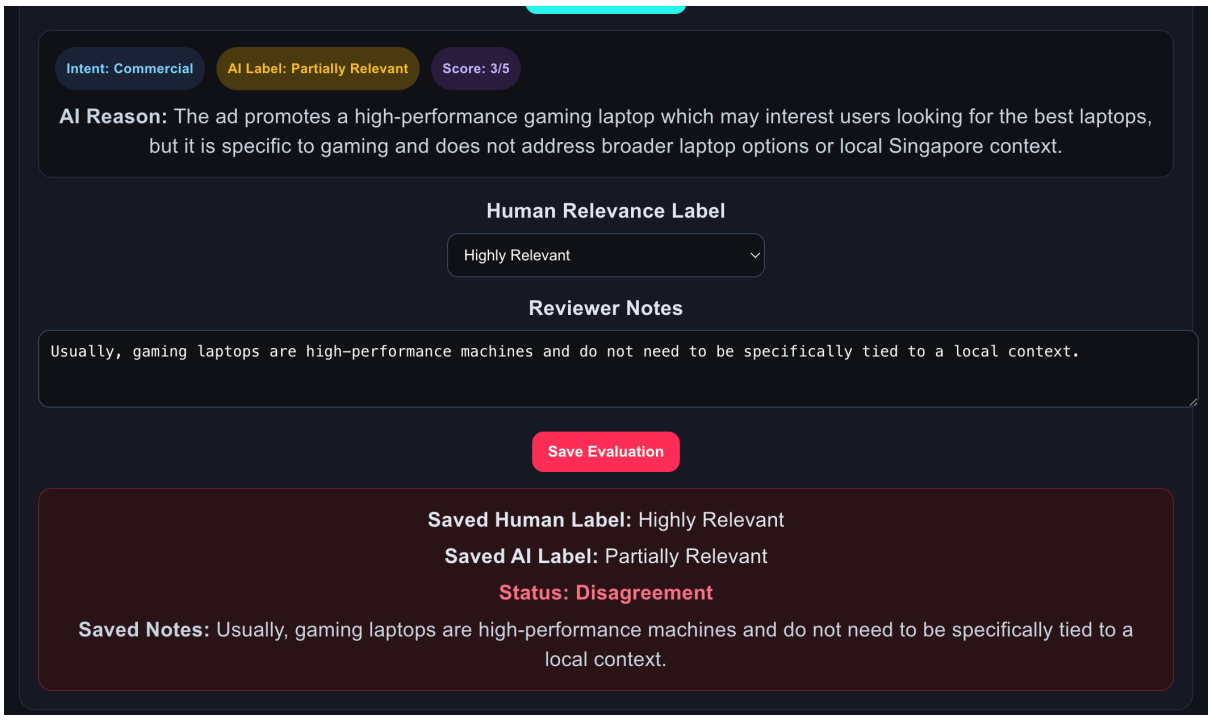
🔷 AI Evaluation Output



Description:

- Shows intent, label, score, reasoning
- Acts as first-pass decision

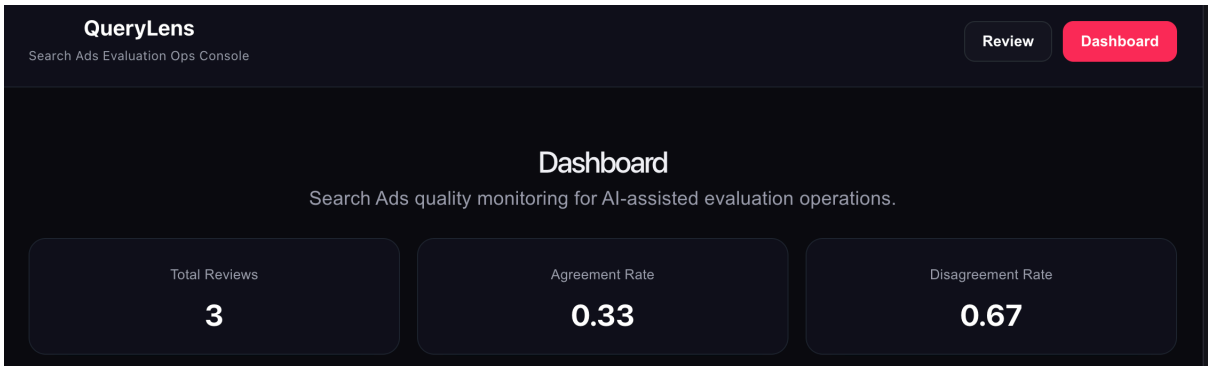
🔷 Disagreement Highlight



Description:

- Shows AI vs human mismatch
- Flags potential model errors

🔷 Dashboard Overview



Description:

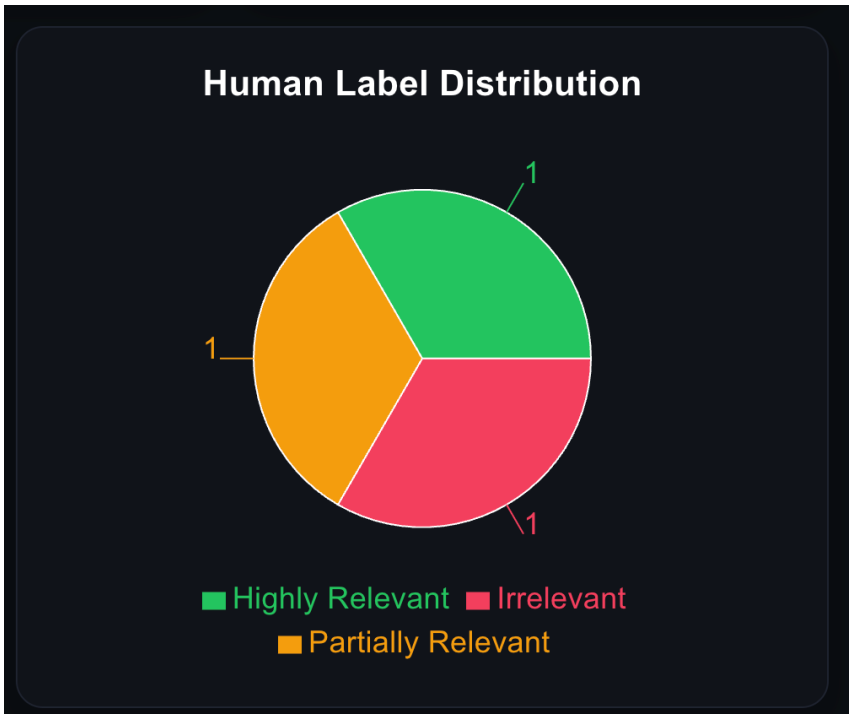
- total reviews
- agreement rate
- disagreement rate

🔷 Query-Level Disagreement Chart

Description:

- identifies problematic queries
- helps prioritize improvements

🔷 Label Distribution Chart



Description:

- shows how ads are being labelled
- indicates quality trends

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8. Tech Stack

Frontend:

- React (Vite)
- Axios
- Recharts

Backend:

- FastAPI
- SQLAlchemy
- SQLite

AI:

- OpenAI API

Deployment:

- Render-ready architecture

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9. Outcome

This MVP demonstrates:

- end-to-end evaluation workflow
- AI-assisted decision making
- human override logic
- disagreement analysis
- metrics-driven insights

It reflects how a real operations tool could support:

- quality control
- model improvement
- workflow efficiency

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10. What I Would Improve Next

- add authentication and user roles
- migrate to Postgres
- support bulk case ingestion
- version labeling guidelines
- export disagreement datasets
- introduce task queues for reviewers

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11. Key Takeaway

This project shows the ability to:

- translate a business problem into a product workflow
- design human-AI collaboration systems
- build measurable evaluation processes

It demonstrates not just building AI features, but designing systems around how AI is used in real operations.

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12. One-line Summary

QueryLens transforms search ads evaluation from a manual review task into a **data-driven, AI-assisted workflow with measurable quality insights**.